THE FORCE™ PLATFORM ACHIEVES ROBUST KNOCKDOWN OF TOXIC HUMAN NUCLEAR *DMPK* RNA AND FOCI REDUCTION IN DM1 CELLS AND IN NEWLY DEVELOPED hTfR1/DMSXL MOUSE MODEL

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BACKGROUND

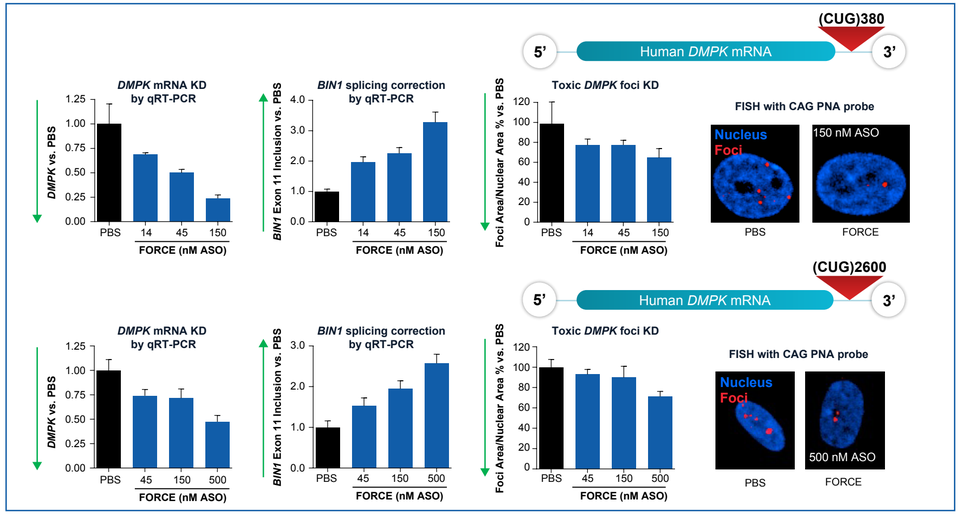
- Myotonic dystrophy type 1 (DM1) is a rare, debilitating, genetic, progressive neuromuscular disease caused by expansion of CUG repeats in the 3' untranslated region of the dystrophia myotonica protein kinase (*DMPK*) RNA[1]
  - *DMPK* transcripts with CUG repeats expansion are trapped in the nucleus and bind to muscleblind-like (MBNL) splicing factors, sequestering them in toxic nuclear foci,[2] ultimately resulting in splicing defects[3]
  - Currently, there are no approved therapies for DM1[4]
- To address the genetic basis of DM1, we designed a FORCE conjugate to target the *DMPK* RNA for RNAse-H-mediated degradation by an antisense oligonucleotide (ASO). The ASO is joined by a clinically validated valine-citrulline linker to an antigen- binding fragment (Fab) antibody that targets the human transferrin receptor 1 (hTfR1), which is highly expressed on muscle
- Using DM1 patient-derived cells we report that a fully human FORCE conjugate:
  - Leads to *DMPK* knockdown (KD)
  - Corrects *BIN1* splicing
  - Reduces toxic *DMPK* nuclear foci
- In an innovative *in vivo* DM1 model developed by Dyne (hTfR1/DMSXL mice), we demonstrated that our FORCE conjugate:
  - Targets the genetic driver of DM1, namely human toxic *DMPK*, in cardiac and skeletal muscle
  - Reduces toxic human *DMPK* foci in the heart
  - Leads to substantial toxic human *DMPK* KD with low and infrequent dosing

METHODS

- DM1 patient myotubes containing short (380) and long (2600) CUG repeats in the *DMPK* RNA were used for *in vitro* analyses
- *DMPK* RNA KD and foci reduction were assessed in a novel hTfR1/ DMSXL mouse model that expresses hTfR1 and human *DMPK* RNA containing > 1000 CUG repeats, representative of a severe DM1 phenotype (DMSXL mouse described previously[2])

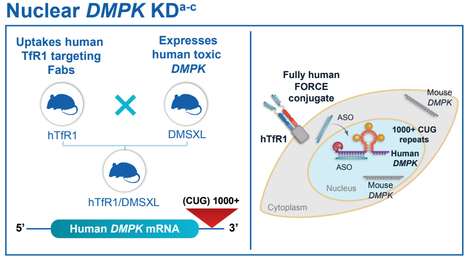
RESULTS

Figure 1. FORCE Conjugate Demonstrated Dose-dependent *DMPK* KD, Splicing Correction, and Foci Reduction in DM1 Myotubes with 380 and 2600 CTG Repeats



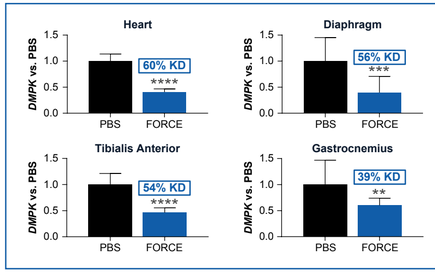
Data are mean ± SD; n = 4. ASO, antisense oligonucleotide; *DMPK*, dystrophia myotonica protein kinase; DM1, myotonic dystrophy type 1; FISH, fluorescence *in situ* hybridization; KD, knockdown; PBS, phosphate-buffered saline; PNA, peptide nucleic acid; qRT-PCR, quantitative reverse-transcriptase polymerase chain reaction.

Figure 2. The hTfR1/DMSXL Mouse Innovative Model Evaluated FORCE Conjugate Pharmacodynamics by Measuring Toxic Human Nuclear *DMPK* KD[a-c]



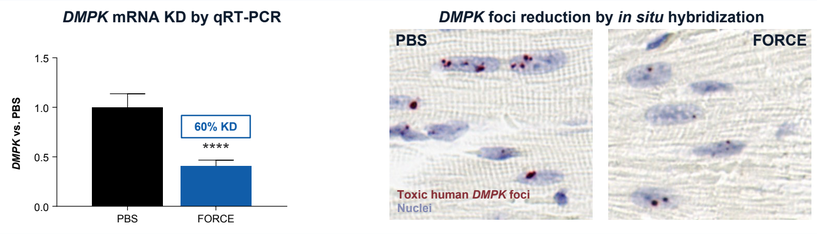
[a]DMSXL mice were first described in *PLoS Genetics* 2012, 8(11):e1003043.
[b]Expresses human TfR1 receptor, enabling use of human TfR1-targeting Fabs.
[c]Underestimates potency, expressing >10 times less human toxic *DMPK* vs mouse *DMPK*. ASO, antisense oligonucleotide; *DMPK*, dystrophia myotonica protein kinase; Fabs, antigen-binding fragments; KD, knockdown; TfR1, transferrin receptor 1.

Figure 3. Repeat Dose of FORCE Conjugate Demonstrated Robust Toxic Human *DMPK* KD in hTfR1/DMSXL Mice after 14 Days



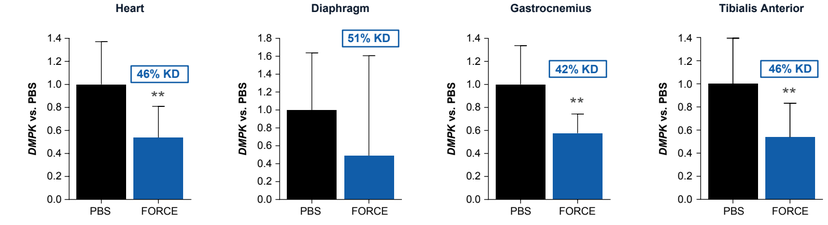
2 × 10 mg/kg, day 0 and day 7, analyzed day 14; Data are mean ± SD; **P* < 0.05, ***P* < 0.01, ****P* < 0.001, *****P* < 0.0001, significant by *t* test.
DMPK, dystrophia myotonica protein kinase; KD, knockdown; PBS, phosphate-buffered saline.

Figure 4. FORCE Conjugate Demonstrated Robust Toxic Human *DMPK* mRNA and Foci KD in the Heart of hTfR1/DMSXL Mice



10 mg/kg ASO day 0 and day 7, analyzed day 14; data are mean ± SD; n = 6–9; *****P* < 0.0001, significant by *t* test.
ASO, antisense oligonucleotide; *DMPK*, dystrophia myotonica protein kinase; KD, knockdown; qRT-PCR, quantitative reverse-transcriptase polymerase chain reaction.

Figure 5. Single Dose of FORCE Conjugate Achieved Sustained Toxic Human *DMPK* KD at Week 4 in hTfR1/DMSXL Mice



10 mg/kg ASO day 0, analyzed day 28; data are mean ± SD; n = 6; ***P* < 0.01, significant by *t* test.
ASO, antisense oligonucleotide; *DMPK*, dystrophia myotonica protein kinase; KD, knockdown; PBS, phosphate-buffered saline.

CONCLUSIONS

- We designed a FORCE conjugate to address the genetic basis of DM1 by targeting toxic nuclear *DMPK* RNA
- These new data demonstrated that our FORCE conjugate can:
  - Correct the DM1 phenotype of patient-derived myoblast cultures with a range of repeats, including those representative of severe DM1
  - Reduce toxic nuclear human *DMPK* foci in cardiac muscle of hTfR1/DMSXL mice
  - Lead to sustained KD of toxic nuclear human *DMPK* in hTfR1/DMSXL mice after a single dose
- These data strongly support further development of our FORCE conjugate, including a planned clinical study

REFERENCES

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2. Huguet A, et al. *PLoS Genet*. 2012;8:e1003043. doi:10.1371/journal.pgen.1003043
3. Nakamori M, et al. *Ann Neurol*. 2013;74:862–872.
4. Thornton CA, et al. *Curr Opin Genet Dev*. 2017;44:135–140.

DISCLOSURE INFORMATION

All authors are employees or advisors of Dyne Therapeutics Inc and may hold Dyne stock and/or stock options.

ACKNOWLEDGMENT
Supported by funding from Dyne Therapeutics Inc.
We thank Dr Geneviève Gourdon for providing the DMSXL mice.
Editorial assistance, funded by Dyne Therapeutics, provided by Alicia Salinero, PhD, of JB Ashtin.

Presented at the Myotonic Dystrophy Foundation 2021 Annual Conference, Virtual, September 10–11, 2021.
Previously presented at the American Society of Gene and Cell Therapy Annual Meeting, Virtual, May 11–14, 2021.